Jay Rao

(475) 225-5010 | jayrao@u.northwestern.edu | linkedin | github | jayrao.me

EDUCATION

Northwestern University, Evanston, IL *Expected June 2028*

Bachelor of Arts (B.A.) in Computer Science, Concentration in Artificial Intelligence

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Scalable Software Architecture, Discrete Math

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Frameworks & Libraries: FastAPI, Node.js/Express, React, PyTorch, CatBoost, NumPy, Pandas, Playwright, pytest

Tools & Platforms: Git/GitHub, GitHub Actions (CI/CD), AWS (EC2, ECS, VPC, ALB, RDS, S3, SES, IAM, Secrets Manager), Terraform, Docker, PostgreSQL, MySQL, Linux/Bash, Agile/Scrum

EXPERIENCE

Software Engineering Intern | Lastrada Music Technology, Inc. *Sep. 2026 – Present*

- Built and integrated a TypeScript intake agent into a music-clearance platform, combining deterministic parsing with Claude to extract details from emails and PDF/DOCX attachments at 99% field accuracy across 12 labeled evaluation documents
- Deployed the agent to AWS using CDK, with Lambda processing, S3 document storage, and DynamoDB thread state; implemented duplicate detection, concurrency control, failure retries, and CloudWatch alarms
- Scoping turnkey third-party product integrations and a Roam-based virtual office for multi-agent workflow orchestration

Founding Engineer | GridGate LLC *Feb. 2026 – Present*

- Sole engineer on GridGate Queue Analysis v1.0, an end-to-end data platform built around a 19-source data lake fed by 20+ scrapers, a 9-module feature pipeline, and a modeling layer covering 3,700+ MISO interconnection projects
- Automated ingestion of 300,000+ state and county regulatory filings and meeting minute PDFs across 15 states and 1,000+ counties using fault-tolerant scrapers with retry logic plus LLM-driven discovery for non-standard sites
- Trained and validated two CatBoost classifiers covering project success and withdrawal (0.93 ROC-AUC, 85% accuracy) and 5-class withdrawal phase (0.94 one-vs-rest ROC-AUC), each with SHAP explainability for stakeholders
- Designed a 6-tier entity resolution pipeline that maps 3,700+ interconnection projects to their parent developers at 83% precision using document-grounded lookups and multi-stage evidence verification against public filings

Technology Manager | iloveyou LLC *Feb. 2026 – Sep. 2026*

- Lead developer managing a team of 3 junior engineers through sprint planning, task assignment and code review, directing backend development and an AI-assisted iOS build from an empty repository to a shipped v1.0 production release
- Built the backend from scratch in Node.js/Express on PostgreSQL with JWT auth and refresh token rotation, email and phone verification (Amazon SES, Twilio Verify), APNs push notifications and REST APIs
- Engineered AWS infrastructure as code in Terraform (VPC, ECS Fargate, ALB, RDS, S3, CloudFront, SES, Secrets Manager, IAM) with remote state and DynamoDB locking, plus CI/CD pipelines in GitHub Actions

PROJECTS

game.sense | Python, PyTorch, PyTorch Geometric, FastAPI, PyQt6, pytest

- Building a Rocket League trajectory-prediction engine using a 4-layer GATv2 encoder, per-entity LSTMs, and a Gaussian mixture decoder to forecast ball and car positions up to 5 seconds ahead from a single player's masked viewpoint
- Reduced one-second ball position error by 56% versus a simple physics baseline and hidden-car error by 60% versus last-seen estimates across 218 validation replays comprising 42,046 windows
- Built a PyQt6 replay-analysis workbench backed by FastAPI/WebSockets, with Windows-to-WSL replay imports, synchronized playback and overlays, forecast inspection, and JSON/CSV exports

nuPython Interpreter & Debugger | C, C++, STL, Valgrind

- Built a Python-like interpreter in 2,000+ lines of C/C++ with recursive expression evaluation, int/float coercion, control flow, runtime type checks, and leak-free memory management verified under Valgrind
- Implemented a GDB-style source-level debugger that unlinks and relinks the program graph to single-step execution, with support for live variable inspection, memory dumps and full execution history tracking

vibes.fm | React, Node.js, Express, MySQL, AWS, Transformers.js, OAuth 2.0

- Engineered a full-stack playlist generator that embeds track metadata, audio features and lyrics into a 384-dimensional vector, then ranks an in-memory cache by cosine similarity and exports matches to Spotify via PKCE OAuth 2.0, deployed on EC2 behind nginx